Credit-risk monitor — architecture

Flow raw CSVs → ingestion (pandas) → analysis panel (parquet) → analytics facts JSON
→ sklearn LogisticRegression + LightGBM → scores parquet
→ IsolationForest anomaly list → Streamlit dashboard (app/app.py).

Tools by stage

Ingestion: pandas, pyarrow. Gender M/F mapped; missing employment tenure / dependents / residence years / interest rate / bureau DPD imputed; rows kept unless target or amount unusable.

Join grain: one row per loan. loans ■ customers (customer_id) ■ bureau agg ■ repay agg ■ collections agg.

Analytics: pandas group-bys written to data/processed/facts.json (the only numbers the LLM may use).

Model: sklearn pipeline (OneHot + scaler + logistic, class_weight=balanced) and LightGBM on bureau, repayment (max/mean DPD, missed EMIs, paid ratio) and collections counts. joblib in models/.

Anomaly: IsolationForest (contamination 3%) plus peer z-scores (loan_type × region).

Dashboard: Streamlit + Plotly. Filters apply to every view.

Ask the Portfolio (LLM)

Called from the Ask tab when OPENAI_API_KEY is set. Payload: the question plus a slim facts.json (portfolio totals and grouped rates only). No account-level rows, names, or IDs leave the app.

If the key is missing, a local retriever answers from the same facts and still cites the grouping.

Unknown questions are declined. Temperature 0.

Anomaly detection

200 accounts flagged into a review queue. Analyst confirm/dismiss is stored locally.

Host: Streamlit process; optional OpenAI Chat Completions (gpt-4o-mini by default).